

Week 3 Final Project Report: MLFQ Scheduler with Priority Boosting

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1. Executive Summary

This report documents the completion of the Multi-Level Feedback Queue (MLFQ) scheduler implementation in xv6. Week 3 focused on implementing priority boosting to prevent starvation, enhancing time quantum enforcement, and validating fairness between CPU-bound and I/O-bound processes. The final implementation includes all core MLFQ features with comprehensive testing and validation.

2. Priority Boosting Implementation

2.1 Starvation Prevention Mechanism

Problem: CPU-bound processes could remain in low-priority queues indefinitely

Solution: Periodic priority boosting moves all processes to the highest priority queue

Implementation Details:

```
#define BOOST_INTERVAL 1000 // Boost every 1000 ticks
```

```
void mlfq_boost_all(void) {  
    // Move all RUNNABLE processes to queue 0  
    for(p = proc; p < &proc[NPROC]; p++) {  
        if(p->state == RUNNABLE && p->priority > 0) {  
            mlfq_enqueue(p, 0);  
        }  
    }  
}
```

2.2 Boost Timer Management

- Global boost timer increments every tick
- When timer reaches BOOST_INTERVAL (1000 ticks):
 - All processes moved to Queue 0 (highest priority)
 - Time tracking counters reset
 - Boost timer reset to 0

3. Enhanced Time Quantum Enforcement

3.1 Time Slice Tracking

Process Structure Enhancements:

```
struct proc {  
    int priority;      // Current queue (0-3)  
    int time_slice_used; // Ticks used in current slice  
    int ticks_in_queue; // Total ticks in current queue  
};
```

Timer Interrupt Handling (kernel/trap.c):

```
if(which_dev == 2) { // Timer interrupt  
    struct proc *p = myproc();  
    if(p && p->state == RUNNING) {  
        p->time_slice_used++;  
        p->ticks_in_queue++;  
    }  
}
```

3.2 Demotion Logic

Scheduler Decision Making:

```
if(p->time_slice_used >= mlfq.time_quantum[queue]) {  
    // Demote to next lower queue  
    int new_queue = (queue < NQUEUES - 1) ? queue + 1 : queue;  
    mlfq_enqueue(p, new_queue);  
    printf("MLFQ: PID %d demoted from Q%d to Q%d\n",  
        p->pid, queue, new_queue);  
} else {  
    // Keep in same queue  
    mlfq_enqueue(p, queue);  
}
```

4. boostproc System Call (Optional)

4.1 Manual Priority Control

Implemented boostproc() system call for testing and manual intervention:

System Call Implementation:

```
int boost_process(int pid) {  
    // Boost specified process to queue 0  
    mlfq_enqueue(p, 0);  
    printf("MLFQ: Manually boosted PID %d to Q0\n", pid);  
}
```

User Interface:

```
int boostproc(int pid); // User-space declaration
```

5. Comprehensive Testing Strategy

5.1 Test Suite Components

final_test.c - Fairness Validation:

// Tests CPU-bound vs I/O-bound process fairness

CPU-bound: Heavy computation, should get demoted

I/O-bound: Frequent yielding, should maintain priority

long_test.c - Extended Behavior Testing:

// Long-running process to observe multiple scheduling events

10 phases of CPU-intensive work

Observes demotion and boosting behavior

mlfq_monitor.c - Concurrent Execution Testing:

// Launches multiple process types simultaneously

Monitors system behavior under mixed workload

5.2 Testing Methodology

Individual Process Testing:

- Verify each process type completes successfully
- Confirm proper priority queue behavior

Concurrent Execution Testing:

- Multiple processes running simultaneously
- Background/foreground process management
- System responsiveness validation

Fairness Validation:

- CPU-bound vs I/O-bound process comparison
- Starvation prevention verification
- Priority boosting effectiveness

6. Final Demo Results**6.1 Key Test Outcomes****CPU-bound Process Behavior:**

- Starts in Queue 0 (highest priority)
- Uses full 1-tick quantum → Demoted to Queue 1
- Continues execution in lower priority queues
- Eventually boosted back to Queue 0

I/O-bound Process Behavior:

- Starts in Queue 0
- Frequently yields before time quantum expires
- Maintains higher priority levels
- Responsive execution throughout

System Performance:

- All processes complete successfully
- No process starvation observed
- System remains responsive during heavy loads
- Priority boosting occurs periodically

6.2 Observed MLFQ Events**During Final Test Execution:**

text

MLFQ: PID 4 demoted from Q0 to Q1 // CPU-bound process demotion

MLFQ: Priority boost triggered! // Automatic boosting

I/O-bound: working 0/5 → 4/5 // I/O-bound progress

CPU-bound finished // CPU-bound completion

I/O-bound finished // I/O-bound completion

7. Final MLFQ Specification

7.1 Queue Configuration

Queue Level	Time Quantum	Purpose
Queue 0	1 tick	Highest priority, interactive tasks
Queue 1	2 ticks	High-responsiveness tasks
Queue 2	4 ticks	Normal applications
Queue 3	8 ticks	CPU-intensive background tasks

7.2 Scheduling Policies

Process Admission:

- New processes start in Queue 0
- Forked children inherit parent's MLFQ state

Queue Promotion:

- Automatic priority boosting every 1000 ticks
- Manual boosting via boostproc system call

Queue Demotion:

- Process uses full time quantum → Demote one level
- Maximum demotion to Queue 3 (no further demotion)

Time Slice Management:

- Time slices enforced per queue level
- Processes yielding early maintain priority

- Time counters reset on queue changes

8. Code Quality and Maintenance

8.1 Implementation Quality

- **Clean Architecture:** MLFQ logic separated from core scheduler
- **Proper Locking:** Thread-safe queue operations
- **Error Handling:** Graceful handling of edge cases
- **Performance:** Minimal scheduler overhead

8.2 Maintainability

- **Modular Design:** Easy to modify queue counts or time quantum
- **Debug Support:** Strategic logging for behavior observation
- **Testing Infrastructure:** Comprehensive test suite
- **Documentation:** Clear code comments and structure

9. Screenshots Section

```
qemu: terminated
faraz@faraz-VirtualBox:~/Projects/xv6-mlfq-week1$ make qemu
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ final_test
=== Final MLFQ Test ===
CPU-bound (PID 4) starting in I/O-bound (PID 5) starting in Q0
I/O-bound: working 0/5
Q0
MLFQ: PID 4 demoted from Q0 to Q1
CPU-bound finished
I/O-bound: working 1/5
I/O-bound: working 2/5
I/O-bound: working 3/5
I/O-bound: working 4/5
I/O-bound finished
Final test completed
$ long_test
MLFQ: Priority boost triggered!
Long test PID 6: Starting
Phase 0/10: Starting CPU work
MLFQ: PID 6 demoted from Q0 to Q1
Phase 0/10: CPU work done
Phase 1/10: Starting CPU work
Phase 1/10: CPU work done
Phase 2/10: Starting CPU work
Phase 2/10: CPU work done
Phase 3/10: Starting CPU work
Phase 3/10: CPU work done
Phase 4/10: Starting CPU work
Phase 4/10: CPU work done
Phase 5/10: Starting CPU work
Phase 5/10: CPU work done
Phase 6/10: Starting CPU work
Phase 6/10: CPU work done
Phase 7/10: Starting CPU work
Phase 7/10: CPU work done
Phase 8/10: Starting CPU work
Phase 8/10: CPU work done
Phase 9/10: Starting CPU work
Phase 9/10: CPU work done
Long test completed
```

```
Long test completed
$ cpu_hog&
$ CPU Hog PID 8: Starting CPU-intensive work
CPU Hog PID 8: Progress 0/4
CPU Hog PID 8: Progress 1/4
MLFQ: PID 8 demoted from Q0 to Q1
CPU Hog PID 8: Progress 2/4
CPU Hog PID 8: Progress 3/4
MLFQ: Priority boost triggered!
CPU Hog PID 8: Finished with result 820578176
ls
.          1 1 1024
..         1 1 1024
README    2 2 2425
cat        2 3 35496
echo      2 4 34408
forktest  2 5 17368
grep       2 6 42848
init       2 7 34864
kill       2 8 34336
ln         2 9 34144
ls         2 10 41624
mkdir      2 11 34392
rm         2 12 34376
sh         2 13 56216
stressfs   2 14 35256
usertests  2 15 186136
grind      2 16 50608
wc         2 17 36432
zombie     2 18 33760
logstress  2 19 36304
forphan    2 20 35160
dorphan    2 21 34600
cpu_hog    2 22 34568
io_bound   2 23 34440
mixed_workload 2 24 34848
mlfq_monitor 2 25 35744
priority_test 2 26 34736
simple_test 2 27 34056
long_test  2 28 34432
final_test 2 29 34952
testprocinfo 2 30 34480
console    3 31 0
$ MLFQ: Priority boost triggered!
echo "Testing responsiveness"
"Testing responsiveness"
```



```

mlfq_monitor
=== MLFQ Scheduler Monitor ===
This program monitors process behavior under MLFQ

Monitor PID: 11
Launched test processes:
  CPU Hog: PID 12
  I/O Bound: PID 13
  Mixed: PID 14

Monitoring for 50 ticks...
I/O Bound PID 13: Starting I/O simulation
I/O Bound PID 13: Cycle 0 - Doing I/O work
Mixed Workload PID 14: Starting mixed work
Mixed PID 14: Cycle 0 - Starting CPU phase
Mixed PID 14: CPU Hog PID 12: Starting CPMonitor: Tick 0 - System running
U-intensive work
CPU Hog PID 12: Progress 0/4
I/O Bound PID 13: Cycle 0 - I/O complete
I/O Bound PID 13: Cycle 1 - Doing I/O work
Cycle 0 - CPU Hog PID 12: Progress 1/4
PU done, now I/O phase
Monitor:MLFQ: PID 12 demoted from Q0 to Q1
Tick 5 - System running
CPU Hog PID 12: I/O Bound PID 13: Cycle 1 - I/O Monitor: Tick 10 - System running
complete
I/O Bound PID 13: Cycle 2 - Doing I/O work
Progress 2/4
CPU Hog PMixed PID 14: Monitor: Tick 15 - System running
I/O Bound PID 13: Cycle 2 - I/O completeMonitor: Tick 20 - System running
4: Cycle 0 - I/O done
M
I/O Bound PID 13: Cycle 3 - Doing I/O work
I/O Bound PID 14: Cycle 1 - Starting CPMonitor: Tick 25 - System running
I/O Bound PID 13: Cycle 3 - I/O complete
I/O Bound PID 14: Cycle 1 2: Progress 3/4
I/O Bound PID 13: Cycle 4 - Doing I/O work
- CPU done, now I/O phase
CPU Hog PID 12: Finished with result Monitor: Tick 30 - System running
820578176
I/O Bound PID 13: Cycle 4 - I/O completeMonitor: Tick 35 - System running

I/O Bound Mixed PID 14: Cycle 1 - I/O done
Mixed PID 13: Cycle 5 - Doing I/O workMonitor: Tick 40 - System running
work
Mixed PID 14: Cycle 2 - Starting CPU phase
I/O Bound PID 13: Cycle 5 - I/O complete
I/O Bound PID 13: Monitor: Tick 45 - System running

```

```

Monitor: Finished monitoring

```

Terminal - faraz@faraz-Vir...

```
Monitor: Finished monitoring
Mixed PID 14: Cycle 2 - Cycle 6 - Doing I/O work
CPU done, now I/O phMLFQ: Priority boost triggered!
ase
I/O Bound PID 13: Cycle 6 - I/O complete
I/O Bound PID 13: Cycle 7 - Doing I/O work
Mixed PID 14: Cycle 2 - I/O done
Mixed PID 14: CyI/O Bound PID 13: Cycle 7 - I/O complete
I/O Bound PID 13: Cycle 8 - Doing I/O work
cle 3 - Starting CPU phase
MixI/O Bound PID 13: Cycle 8 - I/O complete
I/O Bound PID 13: Cycle 9 - Doing I/O work
ed PID 14: Cycle 3 - CPU done,I/O Bound PID 13: Cycle 9 - I/O complete
I/O Bound PID 13: Finished
now I/O phase
Mixed PID 14: Cycle 3 - I/O done
Mixed PID 14: Cycle 4 - Starting CPU phase
MLFQ: PID 14 demoted from Q0 to Q1
Mixed PID 14: Cycle 4 - CPU done, now I/O phase
Mixed PID 14: Cycle 4 - I/O done
Mixed PID 14: Finished
All test processes completed
```

10. Conclusion

10.1 Project Success Metrics

- ✅ **Complete MLFQ Implementation:** All core features working
- ✅ **Starvation Prevention:** Priority boosting effective
- ✅ **Time Quantum Enforcement:** Proper demotion behavior
- ✅ **Fair Scheduling:** Both process types complete successfully
- ✅ **System Stability:** No crashes or deadlocks observed
- ✅ **Performance:** Minimal overhead, maintained responsiveness

10.2 Technical Achievements

Algorithm Implementation:

- Faithful MLFQ scheduler implementation
- Proper multi-level queue management
- Effective time slice enforcement
- Robust priority boosting mechanism

System Integration:

- Seamless integration with xv6 architecture
- Maintained all existing system functionality

- Added comprehensive testing infrastructure
- Implemented optional system call for flexibility

Quality Assurance:

- Extensive testing under various workloads
- Validation of fairness and starvation prevention
- Performance benchmarking and optimization
- Code quality and maintainability focus

10.3 Academic and Practical Value

This implementation demonstrates core operating system concepts:

- **Process Scheduling:** MLFQ as a real-world scheduler
- **Resource Management:** Fair CPU time allocation
- **System Design:** Kernel-level feature implementation
- **Performance Analysis:** Scheduler behavior observation

The completed MLFQ scheduler provides a robust foundation for process scheduling in xv6, balancing responsiveness for interactive tasks with fairness for CPU-intensive workloads.